Meesala Venkatesh

ASIC PHYSICAL DESIGN · VLSI

Vijayawada, Andhra Pradesh, India · mvenky0236@gmail.com
venky-236.github.io · github.com/Venky-236
linkedin.com/in/venkatesh-meesala-46893b330

Electrical & Electronics Engineering graduate (2025) focused on ASIC physical design. I designed my own 41-module electric-vehicle safety SoC and took it from RTL to a routed, DRC-clean layout on SkyWater 130 nm with an open-source flow, after training on Synopsys IC Compiler II. Looking for a Physical Design engineer role and a Master's in semiconductors.

PROJECTS

IVCU-EV V4 — Dual-mode EV safety control SoC · independent design — Jan – Aug 2026

Verilog-2001 · Yosys + ABC · OpenSTA · OpenRAM · OpenROAD · SkyWater SKY130 HD · github.com/Venky-236/ivcu-ev-v4

- Architected a 41-module SoC with 4 asynchronous clock domains (10/25/50/50 MHz), 64 sensor channels, a power-isolated high-voltage safety island, and a 512 × 32 OpenRAM fault log.
- Spent about two months re-architecting the V3 RTL until it synthesized clean: **1,344 → 237** top-level pins via a time-multiplexed ADC interface, **42 → 0** dividers, synchronisers on every clock crossing; standard-cell area **1.19 → 0.32 mm²** (38,238 instances).
- Ran floorplan, power grid, placement, CTS and routing in OpenROAD on a 1230 × 1060 µm die: **55,381** instances at 61% utilisation, **0** global-routing overflow (also under 30% track derating), **0** detailed-routing DRC violations (from 28,340 in 9 passes).
- Timing after CTS (estimated parasitics): worst setup slack +7.951 ns, worst hold slack +0.045 ns, zero setup and hold violators. Traced a −1.094 ns hold failure to an input-delay constraint, fixed the constraint and closed the rest with 302 buffers (+0.2% area, zero setup cost).
- Debugged a power grid reporting 100% of shapes unconnected: 151 metal-1 rails drawn through the SRAM blockage; cutting rows around the macro fixed it.

Guardian Band — women's safety wearable SoC · in progress — Oct – Dec 2025

- Concept and RTL (tap detector, motion monitor, power manager, GPS and cellular blocks), taken through Synopsys Design Compiler and IC Compiler II on a 14 nm teaching library, with clock gating.

Farmer Safety System — 2023 – Jan 2025

- AI snake detection (PIR, thermal, sound + TensorFlow Lite on Raspberry Pi 4) and bite alert; solar powered. My idea; built by a two-person team. **1st place**, Engineers' Day 2023; **1st prize**, College Hackathon 2024. Published (below).

Smart Railway Junction — Jan – Feb 2025

- Arduino junction that switches itself in < 0.5 s and locks instead of routing a train into an occupied loop; tested over 50+ cycles. **1st prize**, Electrical Project Expo, SUNRISE 2K25 (national-level fest).

TRAINING

VLSI First, Hyderabad — Physical Design programme — Feb – Aug 2025 (certified); project work to Oct 2025

- Synopsys IC Compiler II flow: floorplanning, power planning, placement, CTS, routing, timing closure, DRC/LVS verification, Tcl.
- Course designs: **ORCA 14 nm** (32 macros, 72K standard cells, full flow incl. DRC/LVS) and **Chip Top 32 nm** (4 macros, 17K cells, timing closed at 416 MHz through iterative ECO).

Maven Silicon, Bengaluru — VLSI fundamentals — 2 months, 2025

EDUCATION & PUBLICATION

B.Tech, Electrical & Electronics Engineering — NRI Institute of Technology, Vijayawada — Nov 2021 – Apr 2025

First Class, CGPA 7.672 / 10. Coursework: Digital Electronics, VLSI Design, CMOS Technology, Electronic Devices & Circuits.

"Farmer's Safety from Reptiles", *International Journal of Scientific Research and Engineering Development*, Vol. 8, Issue 2, pp. 848–854, 2025. Co-author (5 of 6). doi.org/10.5281/zenodo.15226236

SKILLS

Physical design (intermediate)	Floorplanning, power planning, placement, CTS, routing, timing closure, congestion analysis, STA
EDA tools (intermediate)	Synopsys IC Compiler II, Design Compiler, OpenROAD, OpenSTA, Yosys + ABC, OpenRAM
Digital design	Verilog-2001, clock-domain crossing, clock gating, SDC constraints, SRAM macro integration
Scripting (basic, learning)	Tcl, Python
Other	Linux, WSL, Git & GitHub; Raspberry Pi, Arduino, TensorFlow Lite

OTHER EXPERIENCE

Industrial training, APGENCO Dr. NTTPS thermal power plant (6 weeks, 2023) · Industrial training, Sakura Auto Parts (2 months) · PLC & SCADA automation internship (1 month) · Courses (Udemy, 2026): ASIC Synthesis, STA & PD; UPF Power-Aware Design; Linux & Scripting for VLSI.